Exam Date & Time: 21-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. (CORE STREAM) EXAMINATIONS - NOVEMBER 2024
SUBJECT: ECE 1071 - BASIC ELECTRONICS

BASIC ELECTRONICS [ECE 1071]

Marks: 50 **Duration: 180 mins.**

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Answer all the questions.

1A) A silicon diode has a reverse saturation current of 1pA at 300 K. If the forward voltage across the diode is 0.7 V, find the diode current. If the temperature is increased to 350K, Find the percentage of change in the diode current value. Explain the reason for the temperature dependence of the diode current. (4)

1B) A center-tap full-wave rectifier is connected to a transformer with a turns ratio of 10:1. The input AC voltage to the primary winding of the transformer is 220 V (rms), and the load resistance is 1 kΩ. Assume the diodes are ideal. Find (i) Peak output voltage across the load. (ii) Calculate the average output voltage across the load. (iii)Determine the Peak Inverse Voltage (PIV) rating for each diode. (3)

1C) Explain the construction and operating principle of an LED. (3)

2A) A 5V Zener diode is employed to stabilize the voltage across a 500Ω load resistor when connected to an unregulated power supply fluctuating between 9 and 15 volts. Calculate the suitable range of series resistance values required to ensure the Zener diode operates within its specified current limits of 1 mA to 500 mA. Draw the circuit diagram of the Zener voltage regulator. (4)

2B) Sketch and Explain the I_D-V_G and I_D-V_D characteristics for (a) different channel lengths and (b) Different oxide thicknesses. (3)

2C) A MOSFET device carries 1 mA with $V_{GS} - V_{TH} = 0.6$ V and 1.6 mA with $V_{GS} - V_{TH} = 0.8$ V. If the device operates in the triode region, calculate V_{DS} and W/L. Assume $\mu_n C_{ox} = 200\ \mu A/V^2$ (3)

3A) Draw the circuit diagram of Op-Amp summing amplifier and give the expression for output voltage. Design the circuit using two Op-Amp's to obtain, $V_{out} = 6V_1 - 0.2V_2 - 0.5V_3 + 0.3V_4$, where V_1, V_2, V_3, and V_4 are input voltages. Assume R_F= 10KΩ. Mention the values of the resistances required if the circuit must work as an adder. (4)

3B) In the Op-Amp square wave generator the output saturation voltage is ±12V, R =1kΩ, C = 0.01 μF, R_1 = 1kΩ & R_2 =4kΩ. Draw the circuit. Sketch the voltage waveform across the capacitor and V_0 by marking all the timing and voltage levels. (3)

3C) Perform $(31.75)_{10}$ - $(22.25)_{10}$ using 2's complement method. (3)

4A) Simplify the following logical expression using K-Map and implement using only NAND gates.

F(A,B,C,D)=Σm (0, 2,7, 13, 14, 15)+ Σd (3, 11) (4)

4B) (3)

Represent the given Boolean expression in canonical SOP form and then implement the same using basic gates.

$$f = A\overline{B}C + BC + \overline{(\overline{A}D + A\overline{D})}$$

4C) Design a 3 bit Ripple up counter using positive edge triggered flipflops. Also sketch the timing diagram for the same. (3)

5A) Given an FM signal $V_{FM}(t) = 10 \cos [2\Pi 10^7 t + 5\sin(2\Pi 25000t)]$, Calculate carrier frequency, modulating frequency, frequency deviation and bandwidth using Carson's rule. (4)

5B) For binary data 10110, sketch the ASK, FSK and PSK signals. (3)

5C) Define sampling theorem. Consider an analog signal,

$x(t) = 20\sin(250\prod t) + 5\cos(300\prod t) + \sin(450\prod t)$. (3)

Determine the minimum sampling rate required to reconstruct the signal.

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